Biochemistry Cume

September 18, 2009

1) The melting points of a series of 18-carbon fatty acids are a follows: Steric acid 70° C, Oleic acid 13.5° C, Linoleic acid –5° C, Linolenic acid –11° C.

A) What structural aspect(s) of the 18 carbon units can be correlated with the melting point. (10 pts.)

B) Provide a molecular explanation for the trend in the melting points. (20 pts.)

2) The conductance of a lipid bilayer membrane containing a carrier antibiotic protein decreased abruptly when the temperature was lowered from 40°C to 36° C. In contrast, there was little change in conductance of the same bilayer membrane, over the same temperature range, when it contained a channel-forming antibiotic protein. Give a rational explanation for the observed effects. (20 pts.)

3) Describe what is meant by lipid bilayer membrane "fluidity." Describe one factor that can increase fluidity and one factor that can decrease fluidity. (30 pts.)

4) A common procedure for cleaning grease traps in a sink is to add a product that contains NaOH. Explain why this works. (20 pts.)